

Mayor Climate *Game*

A browser-based prototype for teaching budget--carbon--ventilation tradeoffs.

WHY STUDY THIS?

成本·碳排·通风 如何取舍？

CFD/WRF 太难上手 → 浏览器小游戏能否让权衡可见？

EXPLORATORY PILOT · NOT A REAL-CITY SIMULATION

设施布局 + 事后理解
成本·碳排·通风

假设

一轮游戏后 → 通风题答对·出风口不放煤电

HOW I RAN THE STUDY

One session · logged choices · two survey items

N = 3
SURVEY
志愿者
问卷
Generation AI 2024 · 2026年7
2 题: 预算/电机 + 通风规则

PLANT
游戏
gamebase
2×12 · 100万 · 100万 · 100万 · 100万
能源/历史/耗电/
热泵

Model boundary: rule-based heuristics, not CFD. In-game advisor = deterministic scripts, not live LLM.

PILOT RESULTS · JULY 2026 · DESCRIPTIVE ONLY

Three sessions, two unique layouts

VENTILATION ITEM

3/3

100% answered correctly

BUDGET CHOICE = YES

3/3

All reported coal for cost savings

COAL ON WIND OUTLETS

0/3

No outlet siting in any session

Layout 1 (n=2)



120 t CO₂e ·
score 84

Layout 2 (n=1)



600 t CO₂e ·
score 70

No inferential tests (N = 3). Lower carbon co-occurred with higher score in this pilot.

WHAT IT SUGGESTS

Tradeoffs can be
seen in one
browser round.

支持

通风知识准确

0/3 出风口煤电

张力

3/3 因预算选煤电

碳排 120 vs 600 t

局限

N=3 · 规则模型

探索性 pilot

CLICK INSIDE TO PLAY

Mayor Climate Game



市长气候模拟游戏

你是市长：在 1000 万预算内放置能源设施，观察城市温度、通风和碳排放。

目标：控制碳排放，保持城市通风

第 1 步：在左侧选择一种设施

1 选择设施

预 剩余预算
1000 万元

煤 煤电
80 万 · 碳 120 吨

风 风电
200 万 · 碳 0 吨

泵 地源热泵
160 万 · 碳 15 吨

Demo URL (replace here if you redeploy): <https://mayor-climate-game.vercel.app>

谢谢 *Questions?*

PLAY

mayor-climate-game.vercel.app

PAPER PDF

github.com/HYH-SZ/mayor-climate-game

ADVISOR

Lawted Wu · Generation AI 2026